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FinDiff: Diffusion Models for Financial Tabular Data Generation

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ABSTRACT

The sharing of microdata, such as fund holdings and derivative instruments, by regulatory institutions presents a unique challenge due to strict data confidentiality and privacy regulations. These challenges often hinder the ability of both academics and practitioners to conduct collaborative research effectively. The emergence of generative models, particularly diffusion models, capable of synthesizing data mimicking the underlying distributions of real-world data presents a compelling solution. This work introduces *Financial Tabular Diffusion* (FinDiff), a diffusion model designed to generate real-world mixed-type financial tabular data for a variety of downstream tasks, for example, economic scenario modeling, stress tests, and fraud detection. The model uses embedding encodings to model mixed modality financial data, comprising both categorical and numeric attributes. The performance of *FinDiff* in generating synthetic tabular financial data is evaluated against state-of-the-art baseline models using three real-world financial datasets (including two publicly available datasets and one proprietary dataset). Empirical results demonstrate that *FinDiff* excels in generating synthetic tabular financial data with high fidelity, privacy, and utility.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; **Neural networks**; **Learning latent representations**.

KEYWORDS

neural networks, diffusion models, synthetic data generation, financial tabular data

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1 INTRODUCTION

In the dynamically evolving financial regulatory landscape, data analytics plays an increasingly crucial role. Central banks worldwide, with a particular focus on Europe, amass substantial quantities of microdata. This data is pivotal for guiding policy decisions, conducting risk assessments, and ensuring financial stability. However, the granular nature of this data also engenders unique challenges.

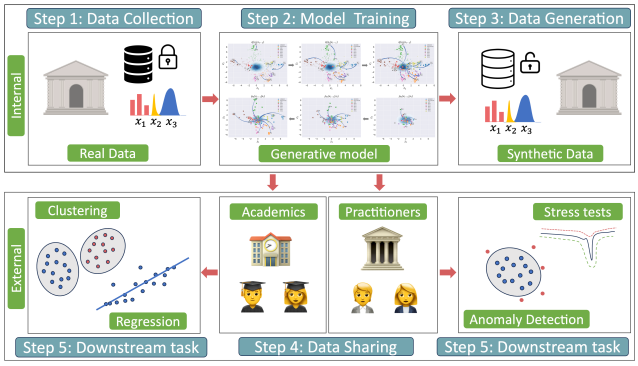

Figure 1: Process of synthetic data generation within the financial regulatory landscape and its subsequent dissemination amongst researchers and practitioners for a variety of downstream tasks, e.g., economic scenario modeling, stress tests, and fraud detection.

At the forefront of these challenges are the stringent regulations surrounding data privacy, such as the *General Data Protection Regulation* (GDPR) in Europe. These regulations inhibit the dissemination of raw financial data, restricting not only independent researchers and practitioners who could employ this data for model training and knowledge discovery but also hindering collaborative research efforts across different central banks.

In response to these challenges, several recent initiatives have underscored the importance of collaborative research. An example is the establishment of the *Financial Big Data Cluster* (FBDC)[1], a collaborative effort bringing together banks, regulatory institutions, Start-ups & SMEs with Research and Development to build a cloud-based financial data pool for developing AI applications. Such clusters could serve as foundations for *Financial Market Authorities* (FMA), e.g. for market manipulation detection or designing AI-based anti-money laundering systems.

Another important aspect is the data sharing, which corresponds to the secure and efficient distribution of data resources across multiple users or organizations, using a mix of technologies and

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[1] https://www.bmwk.de/Redaktion/EN/Artikel/Digital-World/GAIA-X-Use-Cases/financial-big-data-cluster-fbdc.html

legal frameworks, designed to bolster collaboration and risk awareness. The *Irving Fisher Committee's* (IFC) recent guidance note[2] further emphasizes the significance of data sharing and collaboration among central banks. This note explores the challenges, benefits, and initiatives fostering cross-border cooperation, highlighting the potential for harmonizing data practices and leveraging advanced technologies within the financial sector. Furthermore, the *United Nations Economic Commission for Europe* (UNECE) has recently published the guide on data sharing in official statistics[3]. One key aspect to improve economic statistics is to broaden the data scope of national statistical offices beyond their own national data silos by leveraging statistics from international authorities.

In this context, synthetic data generation offers a compelling resolution to these challenges. Synthetic data, mirroring the statistical properties of original data without compromising sensitive information, can broaden the accessibility of essential data for research and machine learning purposes. It facilitates financial model testing, enhances transparency, nurtures collaborative research across institutions, and ensures data privacy regulation compliance. Moreover, synthetic data can simulate diverse economic scenarios, assisting in stress tests and other predictive tasks vital to regulatory operations. Figure 1 depicts such a process with its subsequent scenarios of downstream tasks for researchers and practitioners. One area within AI that holds great promise for finance is generative AI, which encompasses the learning of models to generate new synthetic data with high fidelity [1].

Building upon this motivation, our work introduces a diffusion model designed for synthesizing financial tabular data. Originally developed for image synthesis tasks [5, 40], diffusion models have shown remarkable results in various fields, including natural language processing [41] and audio [4, 17]. Diffusion models simulate a random walk to gradually transform data from a simple initial distribution to the complex distribution of the original data. Due to their unique capabilities to model high-dimensional dependencies, they are well-positioned to address the challenge of data sharing in a regulatory environment. By employing these models, banks can generate synthetic data that maintains the complex statistical properties of the original data, without disclosing any sensitive information. This allows the models to maintain the usefulness of the data for machine learning applications while also ensuring compliance with data privacy regulations.

In summation, this study makes the following contributions:

- The development and implementation of a diffusion model, named *FinDiff*, which capably synthesizes real-world mixed-type financial tabular data.
- The utilization of embedding encoding for addressing the challenges inherent to mixed modality financial data, thus fostering efficient data representation.
- The rigorous assessment of the quality of the data generated by the model using fidelity, privacy, and utility metrics, attesting to the model's precision and effectiveness.

[2]https://www.bis.org/ifc/data_sharing_practices.pdf
[3]https://unece.org/sites/default/files/2021-02/Data%20sharing%20guide%20on%20web_1.pdf

The remainder of this paper is structured as follows: section 2 provides an overview of the related work. In section 3 we describe the diffusion model basics and outline the proposed methodology, *FinDiff*. Next, section 4 and section 5 outline the experimental setup and results. We conclude the paper with a summary and future research directions in section 7. A reference implementation is published via: https://github.com/sattarov/FinDiff.

2 RELATED WORK

The literature survey hereafter focuses on (1) existing Gaussian diffusion models, and (2) developed models designed for the generation of financial data.

2.1 Gaussian Diffusion Models

The initial development of diffusion models can be attributed to Sohl-Dickstein et al. [30] which was followed by a number of improvements and variations like *Denoising Diffusion Probabilistic Models* (DDPM) [13], *Denoising Diffusion Implicit Models* (DDIM) with accelerated sampling) [31]. Another improvement was introduced by Nichol et al. [24] where they compared the coverage of real data by DDPMs and GANs. Rombach et al. showed that with *Latent Diffusion Models* (LDM) [28] one can generate high-quality image synthesis while significantly reducing computation time.

2.2 Discrete Diffusion Models

Strudel et al. [32] introduced *Self-conditioned Embedding Diffusion* (SED), where they successfully showed the applicability of an embedding-based diffusion model for text generation. They also showed that SED can rival autoregressive models. Recently, Gao et al. [10] applied embeddings on the discrete text data generation and showed a number of training challenges such as collapse of the denoising objective or imbalanced scales of the embedding norms. Another example of diffusion models for text generation is the Diffusion-LM introduced by Li et al. [19], however here the authors focused on controlling complex, fine-grained outputs.

2.3 Financial Data Generation

The generation of synthetic financial data has been a topic of interest in recent years. Wiese et al. [35] introduced *Quant GAN*, a data-driven model that utilizes *Temporal Convolutional Networks* (TCNs) to capture long-range dependencies such as the presence of volatility clusters. Ni et al. [22, 23] developed high-fidelity time-series generators, the *SigWGAN*, by combining continuous-time stochastic models with the newly proposed signature W1 metric. Their proposed model was validated on both synthetic data generated by popular quantitative risk models and empirical financial data. Dogariu et al. [6] proposed several solutions for augmenting financial datasets by synthesizing realistic time-series using generative models.

Tabular data is another popular data modality where synthetic data generation is gaining momentum [9]. Here, *Variational Autoencoders* [16] are perhaps one of the most popular models used for tabular data generation [33, 34, 36]. Xu et al. [37] have proposed the *Tabular Variational Autoencoder* (TVAE) specifically designed for tabular data generation tasks. In the same paper, they introduced *Conditional Tabular GAN* (CTGAN), a GAN [12] based model for

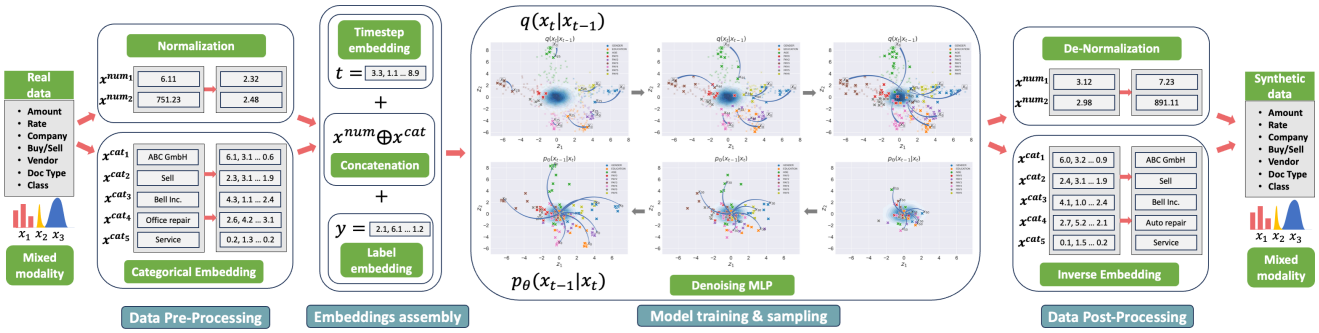


Figure 2: Schematic diagram of the proposed *Financial Tabular Diffusion* (FinDiff) model. The process begins with the pre-processing phase, where categorical attributes are transformed into embeddings and numerical attributes undergo normalization. During the embedding assembly phase, these processed data streams combine into an aggregate $x_0 = e^{c_1} \oplus e^{c_2} \oplus ... \oplus e^{c_k} \oplus x^{num}$, further enriched with time and label embeddings. This combined data is then passed into a Feed Forward Neural Network to estimate the added noise component. Once the model is trained and new data is generated via Gaussian sampling, the post-processing phase maps the resultant embedding $x^{\hat{cat}_i}$ back to its nearest counterpart. Concurrently, numerical attributes are denormalized, restoring to their original data space.

tabular datasets. Lately, a variety of GAN models to generate tabular data have been proposed [7, 8, 29]. The recent attempt to model tabular data using diffusion models was done by Kotelnikov et al. [18]. However, since the categorical attributes were modeled using multinomial diffusion models [14], they had to use one-hot encoding transformation, which has its disadvantage for large datasets. Another recent attempt to model tabular data with diffusion models is *MissDiff* [25] which is capable of training a synthesizer directly on the data with missing values.

To the best of our knowledge, this work presents the first attempt to develop a diffusion model for synthesizing mixed-type financial tabular data using embedding encoding for categorical attributes.

3 METHODOLOGY

In this section, we describe the *Gaussian Diffusion Process* and proposed *Financial Tabular Diffusion* (FinDiff).

3.1 Gaussian Diffusion Models

Denoising diffusion probabilistic model [30], [13] is a latent variable model that utilizes a forward process to perturb the data $x_0 \in \mathbb{R}^d$ step by step with Gaussian noise ϵ, and then restore the data back in the reverse process. The forward process is started at x_0 and latent variables $x_1 ... x_T$ are generated with a Markov Chain by gradually perturbing it into Gaussian noise $x_T \sim \mathcal{N}(0, I)$. Hence, each Markov transition has the following form:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t} x_{x-1}, \beta_t, \mathbf{I}), \tag{1}$$

where β_t is the noise level added at timestep t. Sampling x_t from x_0 for an arbitrary t can also be achieved in a closed form:

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{1-\hat{\beta}_t} x_0, \hat{\beta}_t, \mathbf{I}), \tag{2}$$

where $\hat{\beta}_t = 1 - \prod_{i=0}^{t}(1-\beta_i)$. In the reverse process, the model incrementally denoises the latent variables x_t to recover the data x_0. To approximate this process, we train a neural network with parameters θ and each denoising step is parameterized as:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)), \tag{3}$$

where μ_θ and Σ_θ are the estimated mean and covariance of q. Since Σ_θ is diagonal, as suggested by Ho et al. [13], the computation of μ_θ is the following:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}(x_t - \frac{\beta_t}{\sqrt{1-\hat{\alpha}_t}}\epsilon_\theta(x_t, t)), \tag{4}$$

where $\alpha_t := 1 - \beta_t$, $\hat{\alpha}_t := \prod_{i=0}^{t} \alpha_i$ and $\epsilon_\theta(x_t, t)$ is the predicted noise component. It was empirically shown that simplified loss of mean-squared errors between the ground truth ϵ and estimated $\epsilon(x_t, t)$ empirically leads to better results than tractable variational lower bound $log\ p_\theta(x_0)$:

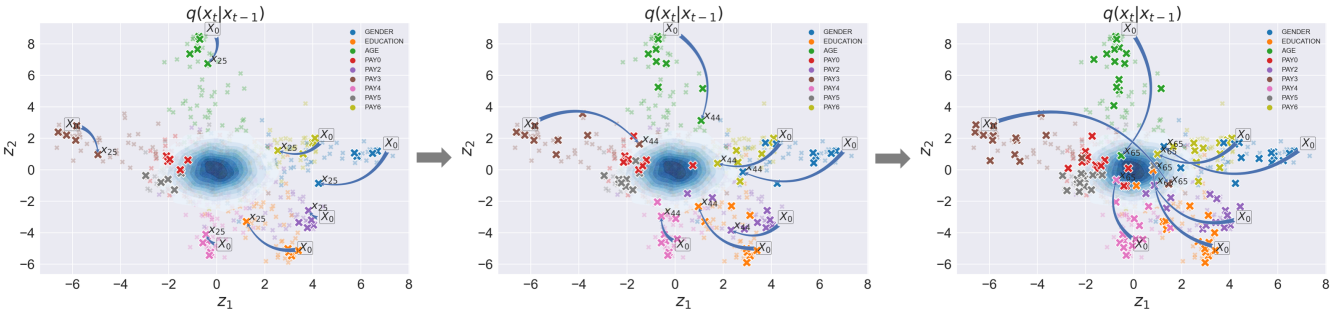
$$\mathcal{L}_t = \mathbb{E}_{x_0,\epsilon,t}||\epsilon - \epsilon_\theta(x_t, t)||_2^2. \tag{5}$$

Though such a framework works well on continuous data, such as images, it cannot be directly applied to discrete data, such as categorical attributes of tabular data.

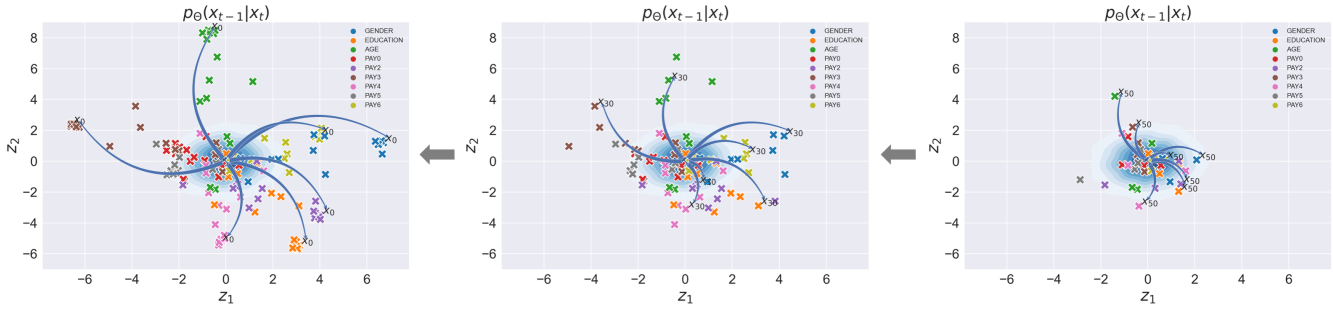
3.2 Financial Mixed-Type Tabular Diffusion

Our approach addresses the challenges inherent to mixed modality financial tabular data, comprising both categorical and numeric attributes. The categorical data, discrete variables exhibiting a finite set of possible values, has often been a stumbling block for conventional numerical models. We transform this categorical data, denoted $x^{cat} = (x^{c_1}, x^{c_2}, ..., x^{c_N})$, into semantically rich, continuous representations via the use of embeddings, $e \in \mathbb{R}^D$. Each categorical element, x^{c_i}, is thus represented in a condensed vector space, positioning similar categories in close proximity, yielding an embedding matrix $E \in \mathbb{R}^{D \times C}$.

As illustrated in Figure 2, the proposed methodology encompasses several stages. In the data pre-processing phase, numeric



(a) The forward diffusion process of three randomly selected noisy batches. This illustration presents the navigational path towards the Gaussian center, produced by gradually perturbing x_0 with Gaussian noise, culminating in the final noisy data representation $x_T \sim \mathcal{N}(0, I)$. In this step, each embedding representation converges towards the center of Gaussian $x_T \sim \mathcal{N}(0, I)$.



(b) Three snapshots of the reverse diffusion process. The model employs this path to drift back towards the initial embedding representation x_0. In this step, the embedding drifts back in the direction of the starting token X_0

Figure 3: Example of the diffusion forward and reverse processes of 9 categorical embeddings of a single observation from the Credit Default dataset. Each cross point symbolizes the 2-dimensional embedding of a specific category, either at its origin x_0 or subsequent noisy representations such as x_{25}, x_{44}, x_{65}. The blurry points symbolize the noisy representations that remain unseen by the model.

attributes undergo normalization while categorical tokens are embedded using a matrix E, which then, in the embedding assembly phase, are concatenated and aggregated with time and label embeddings. Gaussian noise is subsequently introduced according to Equation 2, marking a diffusion step $\mathbf{t}$. To compute the time embedding $\mathbf{t}$, a sinusoidal position embedding is utilized, in alignment with Nichol et al.'s methodology [24]. Both time embeddings and encoded sample x_0 undergo a linear layer before being aggregated, following the approach of Nichol et al. [24]. The representation thus obtained is channeled through a Feed Forward Neural Network, where the model strives to estimate the added noise component, as delineated in Equation 5. Once the model is trained, the sampled new data in the post-processing phase is mapped back to its respective category using the closest representation within the embedding matrix E, while numerical attributes are denormalized.

In Figure 3, the model's training and sampling steps are further described, utilizing a real-world example. This figure depicts the learning trajectory of categorical embeddings from a single observation, alongside the subsequent sampling step. In Figure 3a, the forward process is illustrated, showing the navigation path towards the Gaussian center $\mathcal{N}(0, I)$. During the reverse process, demonstrated in Figure 3b, the model employs this path to drift back toward the initial embedding representation x_0. This arrangement enables the model to perceive the latent structure and interrelations within the categorical features, thereby fostering meaningful correlations during sampling.

4 EXPERIMENTAL SETUP

In this section, the details of the conducted experiments are described, encompassing the data preprocessing, model training, baselines, and evaluation metrics. The *PyTorch* v1.13.0 [26] framework was used to implement the model training and evaluation. The implementations of the evaluation metrics and baseline algorithms are derived from the *Synthetic Data Vault* (SDV) library v1.0.1 [27].

4.1 Datasets and Data Preparation

We evaluate *FinDiff* using three real-world datasets. Below, we provide the description of each dataset:

- **Credit Default**:[4] The dataset comprises bill statements of credit card clients, their default payments, history of payment as well as the demographic factors of the clients during the period April 2005 to September 2005 [38].
- **City of Philadelphia**:[5] The dataset contains checks and direct deposit payments made by the City of Philadelphia during the 2017 fiscal year.
- **Fund Holdings**:[6] This proprietary dataset consists of the individual holdings of investment funds [3]. Each record reflects the asset or liability value submitted by the reporting entity at the end of the month.

Table 1: Descriptive statistics of the selected datasets used to evaluate the proposed *FinDiff* synthetic data generation.

Data	Rows	Columns		Classes
		Categ.	Num.	
Credit Default	30,000	10	13	2
Philadelphia Payments	100,000	7	1	11
Fund Holdings	88,893	6	78	18

In considering dataset selection, we prioritized diversity in the proportion of categorical and numeric attributes. Specifically, the Philadelphia Payments dataset is primarily composed of categorical attributes, the Fund Holdings dataset predominantly contains numeric attributes, and the Credit Default dataset presents a nearly equal distribution of both attribute types, as Table 1 attests. The numeric attributes were transformed with the gaussian Quantile Transformer [7]. Notably, we were constrained to sample only 100,000 observations from the total 238,894 in the Philadelphia dataset due to limitations faced during the training of baseline models. The need for a one-hot encoded representation for these models led to exceedingly high-dimensional transformed datasets of 6,124 (7,830) dimensions. Unlike these baseline models, our proposed method, *FinDiff*, utilizes embeddings instead of one-hot encoding, thereby circumventing such dimensionality issues.

4.2 Diffusion Model Training Setup

Every dataset was split into training and test sets by a fraction of 70 and 30 correspondingly. All models are trained on the train set and the evaluation metrics are collected with respect to the test set.

Architectural Setup. The network architecture utilized for each dataset was selected after the exploration of the hyperparameter space. We determined the most appropriate number of diffusion steps to be 500, using a linear scheduler. The dimensionality of the label and timestep embedding was 64 on all archtiectures. Each categorical attribute was assigned a dimensionality of 2, which proved beneficial for both visual inspection (see Figure 3) and computational efficiency. Moreover, our empirical observations indicated that increasing the dimensionality of the categorical embeddings did not result in a performance enhancement. Strudel et al. [32] similarly reported that a high-dimensional embedding space leads to performance degradation when training such models on textual data.

Table 2: Overview of architectural details used to train the *FinDiff* models on the distinct datasets. Each column reflects the number of neurons per feed-forward layer ℓ.

Data	ℓ=1	2	3	4	5	6
Credit Default	1,024	1,024	1,024	1,024	-	-
Philadelphia Payments	8,192	8,192	8,192	8,192	8,192	8,192
Fund Holdings	1,024	1,024	1,024	1,024	-	-

Hyperparameters. Depending on the dataset, a distinct combination of the number of neurons and layers was selected. The most effective hyperparameters were identified following an exhaustive search across the following space: a number of neurons {256, 512, 1024, 2048, 4096, 8192} and a number of feed-forward layers {4, 6, 8, 10, 12}, resulting in the final network architectures in Table 2. We train every model for a maximum of 3000 epochs with a mini-batch of size 512 and use Adam [15] with $\beta_1 = 0.9$, $\beta_2 = 0.999$ and a cosine learning rate scheduler [20]. The learning weights are randomly initiated as described in [11].

Baselines. To benchmark the performance of the *FinDiff* we compare it against the following three baseline models commonly used in mixed-type tabular synthetic data generation:

- **TVAE** [37] - an adapted version of the variational autoencoder, designed for mixed-type tabular data generation.
- **CTGAN** [37] - GAN-based synthetic tabular data generator, shown to perform well on tabular datasets.
- **TabDDPM** [18] - recently developed diffusion-based alternative for generation of tabular datasets.

4.3 Evaluation Metrics

To assess the quality of generated data, we evaluate the excellence from various viewpoints: fidelity, privacy, utility, and synthesis.

Fidelity. To quantify the relevance of generated data, we evaluate it in terms of fidelity. Fidelity assesses how well the synthetic data mimics the real data. This is performed on the column as well as on the row levels. The column fidelity is examined using the similarity of every column in the synthetic dataset against the real dataset. To measure it for numeric attributes we use Kolmogorov-Smirnov statistic $KS(x^d, s^d)$ [21] which reflects the maximum difference between an empirical and hypothetical cumulative distribution. For categorical attributes, we utilize the Total Variation Distance between the real and synthetic columns, defined as:

$$TVD(x^d, s^d) = \sum_{c \in C} |p(x^{d_c}) - p(s^{d_c})|, \tag{6}$$

where $p(s^{d_c})$ denotes the fraction of existing categories $\mathbf{c}$ in the attribute d. Therefore, the column-wise fidelity (ω_{col}) obtains the following form:

[4]The dataset is publicly available via: https://archive.ics.uci.edu/ml/datasets/default+of+credit+card+clients
[5]The dataset is publicly available via:https://data.phila.gov/visualizations/payments
[6]In compliance with strict data privacy regulations, neither the content nor the descriptive statistics of the dataset can be made publicly available.
[7]https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.QuantileTransformer.html

Table 3: Comparative analysis of different synthetic data generation models on various datasets. Each model is evaluated on several key parameters including Fidelity (Column and Row), Utility, Synthesis, and Privacy. Boldface numbers indicate the best-performing model for a given measure on a particular dataset.

		Evaluation Measures				
Dataset	**Model**	**Fidelity Column ↑**	**Fidelity Row ↑**	**Utility ↑**	**Synthesis ↑**	**Privacy ↓**
Credit Default	TVAE [37]	0.921 ± 0.01	0.924 ± 0.01	0.785 ± 0.01	1.000 ± 0.00	2.550 ± 0.01
	CTGAN [37]	0.884 ± 0.01	0.889 ± 0.01	0.695 ± 0.03	1.000 ± 0.00	2.963 ± 0.07
	TabDDPM [18]	0.382 ± 0.03	0.370 ± 0.01	0.673 ± 0.08	1.000 ± 0.00	3.734 ± 0.18
	FinDiff (ours)	**0.965 ± 0.01**	**0.944 ± 0.01**	**0.795 ± 0.01**	1.000 ± 0.00	**2.118 ± 0.03**
Philadelphia Payments	TVAE [37]	0.791 ± 0.01	0.634 ± 0.01	0.785 ± 0.03	0.992 ± 0.01	2.000 ± 0.00
	CTGAN [37]	0.782 ± 0.01	0.576 ± 0.01	0.728 ± 0.01	0.994 ± 0.01	1.765 ± 0.32
	TabDDPM [18]	0.900 ± 0.01	0.535 ± 0.01	0.863 ± 0.01	**1.000 ± 0.01**	4.135 ± 0.20
	FinDiff (ours)	**0.901 ± 0.01**	**0.838 ± 0.00**	**0.874 ± 0.00**	0.992 ± 0.01	**1.414 ± 0.00**
Fund Holdings	TVAE [37]	0.745 ± 0.01	**0.952 ± 0.01**	0.543 ± 0.02	1.000 ± 0.00	**0.171 ± 0.01**
	CTGAN [37]	0.591 ± 0.02	0.937 ± 0.01	0.421 ± 0.03	1.000 ± 0.00	0.847 ± 0.29
	TabDDPM [18]	0.254 ± 0.01	0.740 ± 0.01	0.215 ± 0.06	1.000 ± 0.00	9.816 ± 8.00
	FinDiff (ours)	**0.829 ± 0.01**	0.949 ± 0.01	**0.581 ± 0.02**	1.000 ± 0.00	3.667 ± 0.40

*Scores are derived from the averaged results and standard deviations of five experiments, each initiated with distinct random seeds

$$\omega_{col}(x^d, s^d) = \begin{cases} 1 - KS(x^d, s^d) & \text{if } d \text{ is num.} \\ 1 - \frac{1}{2}TVD(x^d, s^d) & \text{if } d \text{ is cat.} \end{cases} \tag{7}$$

The total score for the synthetic dataset S is computed as the average across all attributes:

$$\Omega_{col}(X, S) = \frac{1}{D}\sum_{d=0}^{D} \omega(x^d, s^d). \tag{8}$$

The row fidelity is measured using correlations between a pair of columns. For numeric attributes, a Pearson correlation [2] is computed between a pair of columns $\rho(x^a, x^b)$. Next, the absolute difference between the correlations of the real and synthetic attribute pairs $\delta(x^{a,b}, s^{a,b}) = |\rho(x^a, x^b) - \rho(s^a, s^b)|$ is used to estimate the coefficient. For categorical attributes, we used the contingency table by computing the Total Variation Distance on a pair of categories in attributes A and B.

$$\omega_{row}(x^{a,b}, s^{a,b}) = \begin{cases} 1 - \frac{1}{2}\delta(x^{a,b}, s^{a,b}) & \text{if } a,b \text{ are num.} \\ 1 - \frac{1}{2}TVD(x^{a,b}, s^{a,b}) & \text{if } a,b \text{ are cat.} \end{cases} \tag{9}$$

The final fidelity score for a synthetic dataset S is computed as the average score across all attribute pairs:

$$\Omega_{row}(X, S) = \frac{1}{D}\sum_{d=0}^{D} \omega(x^{a,b}, s^{a,b}). \tag{10}$$

Privacy. The privacy metric represents the extent to which the synthetic data prevents identification of the original data entries. To measure the privacy of the generated, we use the *Distance to Closest Records* (DCR). The *DCR* is computed as the closest distance from the synthetic data point s_n to the real data points X, formally defined as:

$$DCR(s_n) = \min_{x \in X} d(s_n, x), \tag{11}$$

where $d(\cdot)$ is the distance metric (Euclidean is our case). The final score is computed as the *DCR* median of all synthetic data points.

Utility Assessment. A critical aspect of synthetic data evaluation is the measurement of utility, essentially quantifying its functional equivalence to the original data. This evaluation involves training a machine learning model on the synthetic dataset, followed by its evaluation on the original dataset. This measurement criterion serves to assess the applicability and quality of the generated data for downstream tasks. We quantify utility by training a classifier on synthetic data of identical dimensions as the training set and subsequently evaluating it on the real test set. The utility is represented by the mean accuracy, aggregated across all classifiers, defined as follows:

$$\Phi = \frac{1}{N}\sum_{i=1}^{N} \Theta_i(S_{train}, X_{test}), \tag{12}$$

where in this study a selection of $N = 5$ classifiers are used. The classifiers include Random Forest, Decision Trees, Logistic Regression, Ada Boost, and Naive Bayes.

Synthesis. Synthesis reflects the ability of a model to generate synthetic data that is not an exact replication of the real data. It checks whether the generated records are novel or exactly match the records in the original dataset. It is computed as the fraction of the matched synthetic records to the total number of generated records [27]. A numeric entry is considered a match if its value is within the 1% range of the real value. This percentage may be adjusted depending on the task and data specificity. The final synthesis score is the complement, defined as follows:

$$\gamma(S) = 1 - \frac{S_{n_m}}{N}, \tag{13}$$

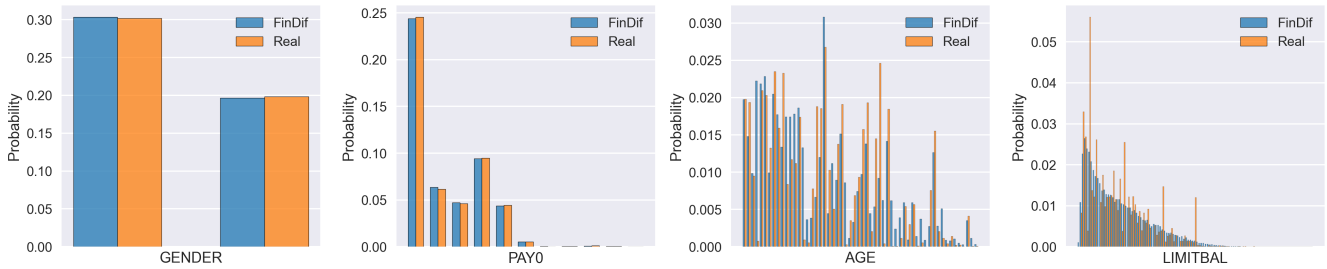


Figure 4: Feature distributions of the real and synthetic data generated by *FinDiff*. The orange bars mimic the real-world data distribution of the Credit Default dataset, and the blue bars are the synthetic data generated by *FinDiff*. Categorical features are 'PAY0', 'AGE' and 'GENDER', the numeric feature is 'LIMITBAL.'

where S_{n_m} is the number of matched synthetic records, and N is the total number of generated records.

5 EXPERIMENTAL RESULTS

In this section, we describe the results of the conducted experiments. We demonstrate the efficiency of the proposed technique, providing quantitative results with the ablation study and qualitative assessment.

5.1 Quantitative Results

To quantitatively evaluate the developed technique, we assess the model from different perspectives. Table 3 shows the results of all metrics across three datasets and all baseline models.

Fidelity. For the Credit Default dataset, the *FinDiff* model outperformed all other models in terms of both column and row fidelity and achieved an approximately 1.2% and 1.6% increase in performance for these metrics compared to the second-best performing model, TVAE. The high fidelity scores suggest that *FinDiff* accurately captures both individual and joint distributions, which is crucial in preserving the integrity of the dataset's statistical properties. On the Philadelphia Payments dataset, the *FinDiff* model again demonstrated superior performance. On the Fund Holdings dataset, the *FinDiff* outperformed all models on column-wise metric and is the second-based model on the row-wise fidelity with a very small margin. We believe, that the reason lies in the nature of the Fund Holding dataset, where most of the numeric attributes are highly skewed with extremely high values.

Privacy. In the context of the Credit Default dataset, the *FinDiff* model establishes a strong performance by yielding the lowest privacy score, measured at 2.118. This outstrips the performances of TVAE, CTGAN, and TabDDPM models by 16.94%, 28.51%, and 43.27% respectively. This finding indicates that *FinDiff* can generate synthetic credit default data while ensuring superior privacy preservation. As for the Philadelphia Payments dataset, *FinDiff* continues its lead, returning the lowest privacy score. Analyzing the Fund Holdings dataset, we encounter an anomalous trend, where the TVAE model records the best privacy score at 0.171. In conclusion, the *FinDiff* model demonstrates robust privacy performance in the Credit Default and Philadelphia Payments datasets, significantly outperforming other models. While it cedes to the TVAE model in the Fund Holdings dataset, it still offers enhanced privacy protection relative to the TabDDPM model. These results underline that the *FinDiff* model generally excels at ensuring privacy preservation when generating synthetic data, although the efficiency of the privacy protection can be influenced by dataset-specific factors.

Utility. Starting with the Credit Default dataset, the *FinDiff* model demonstrates superior utility, with a score of 0.795. This surpasses the performances of TVAE, CTGAN, and TabDDPM models by 1.27%, 14.38%, and 18.12% respectively, indicating that *FinDiff* optimally retains the essential characteristics of the original data. Moreover, when computing the utility accuracy using only real data (trained and tested on the real data), it yields only 0.706 which is 8.4% lower than using the synthetic data of *FinDiff*. This significantly confirms the usefulness and importance of strong synthesizers. For the Philadelphia Payments and Fund Holdings datasets, *FinDiff* continues to demonstrate its leading performance, although exceeding the second-based model by a small margin.

Synthesis. Analyzing synthesis scores across the three datasets, the models display generally strong performances. In the Credit Default and Fund Holdings datasets, all models, including *FinDiff*, achieved perfect scores of 1.000. For the Philadelphia Payments dataset, while the TabDDPM model secured a perfect score, *FinDiff*'s performance was only marginally lower at 0.992.

Upon examining the data from all three datasets, it is evident that *FinDiff* consistently results in high scores across all evaluation measures. It showed high performance in terms of Fidelity (both column and row), with peak scores observed in the Credit Default dataset. Furthermore, *FinDiff* demonstrated admirable privacy and utility numbers, evidenced by the lowest privacy scores for both the Credit Default and Philadelphia Payments datasets. While the model's performance is not the leading one in the Fund Holdings dataset, it still maintains a significant advantage over CTGAN and TabDDPM models. This underscores the generally strong performance of the *FinDiff* model, though the degree of this performance can be contingent upon dataset-specific characteristics.

5.2 Qualitative Results

In order to gauge the quality of the data generated, feature distributions of the original and synthetic data of the *FinDiff* model are plotted, which reflect the column-wise fidelity capabilities. Figure

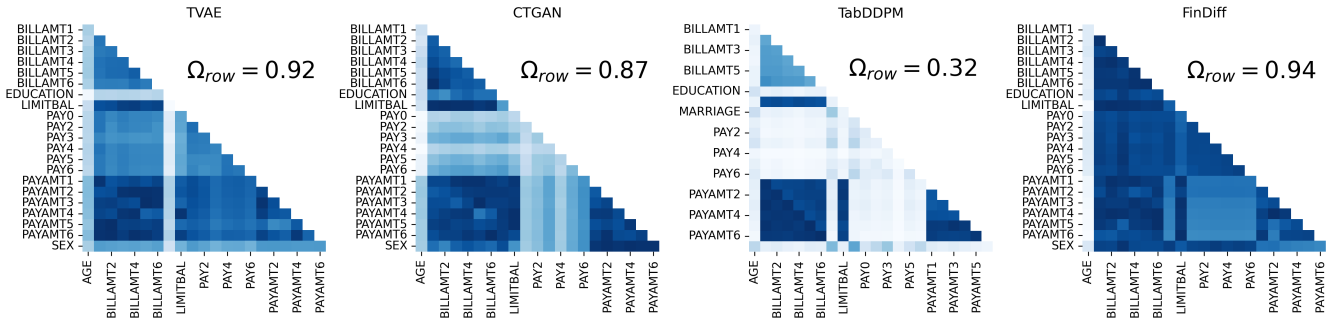


Figure 5: Absolute difference of feature correlations between the Credit Default dataset and its synthetic version generated by all models. A more intensive color gradient indicates a higher correlation between a pair of features and; hence better quality. The mean of all correlation score pairs is located in the top right corner.

4 illustrates examples of three categorical features and a numerical one. In an effort to more effectively assess the quality of sampling, categorical attributes with varying quantities of unique categories are selected, such as 'GENDER'=2, 'PAY0'=11, and 'AGE'=56. The capacity of the *FinDiff* model to faithfully replicate the distributions of the original data is confirmed through this process.

The fidelity level from the row perspective is evaluated by comparing feature correlations of the synthetic data produced by a variety of models. Figure 5 demonstrates an example of such correlations for the Credit Default dataset, spanning all features. A more intense color gradient signifies a stronger correlation between a pair of attributes, implying an enhanced quality of the synthesizer.

6 ABLATION STUDY

We have observed the strong influence of normalization techniques when training the *FinDiff* model applied to numeric attributes of the Fund Holdings. Since this dataset contains mostly extremely skewed numeric attributes, the normalization technique has to be selected carefully. The Table 4 presents a fidelity assessment of three different transformation or scaling techniques applied on Fund Holdings: Standard Scaler [8] (zero mean and unit variance), Power Transformer [9] ('yeo-johnson' method [39]) and Quantile Transformer [10]. Notably, the Quantile Transformer emerges superior, achieving the highest fidelity scores in both metrics. Conversely, the Standard Scaler yields the lowest fidelity, suggesting a less accurate data replication. These results underline the influence of data scaling methods on the effectiveness of synthetic data generation.

7 SUMMARY AND FUTURE WORK

In this work we introduced *FinDiff*, a financial tabular diffusion model, designed for generating mixed-type synthetic financial tabular data aimed at enhancing downstream tasks. The model utilizes embedding encoding to address the challenges inherent to mixed-modality financial data. Furthermore, the model is equipped to generate conditional sampling, proving particularly beneficial for datasets with multiple classes. Evaluations of the model, undertaken from various perspectives including fidelity, privacy, and utility, suggest that the model outperforms all baseline models on public datasets and achieves notable results on proprietary datasets as per the set metrics. The potential for this model to serve as an effective tool within the financial regulatory environment is evident.

The forthcoming research will concentrate on the extension of the *FinDiff* model within the Federated Learning framework. This adaptation aims to produce mixed-type tabular data without necessitating centralized access to primary datasets. Such an advanced model promises to be a pivotal instrument in circumventing the dissemination of confidential data, thus addressing data-sharing challenges for regulatory institutions.

Table 4: Fidelity assessment of various transformation/scaling methods evaluated on the Fund Holdings dataset. The values represent the mean and standard deviation of 5 experiments of random seeds.

	Fidelity Columns	Fidelity Row
Standard Scaler	0.534 ± 0.01	0.824 ± 0.01
Power Transformer	0.552 ± 0.03	0.889 ± 0.04
Quantile Transformer	**0.829 ± 0.01**	**0.949 ± 0.01**

[8] https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html
[9] https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.PowerTransformer.html
[10] https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.QuantileTransformer.html

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